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Using Smartphone Surveys to Predict Next-Week Suicide Attempts

Matthew K. Nock[1,2,3,4], **Evan M. Kleiman**[5], **Kate H. Bentley**[1,2], **Rebecca G. Fortgang**[1,2], **Alexander J. Millner**[1], **Kelly L. Zuromski**[1,3], **Adam Bear**[1], **Alexis Christie**[1,2], **Merryn Daniel**[1,3], **Dylan DeMarco**[1], **Lia Follet**[1,3], **Flynn Kelly**[1,3], **Hope Neveux**[1], **Onyinye Obi-Obasi**[1,2], **Jordyn R. Ricard**[1,2], **Narise Ramlal**[1,2], **Tida Tambedou**[1,2], **Yaniv Yacoby**[1,2,6], **Suzanne A. Bird**[2], **Ralph Buonapane**[3], **Abigail Donovan**[2], **Patrick Mair**[1], **Jukka-Pekka "JP" Onnela**[4], **Rosalind Picard**[7], **Jordan W. Smoller**[2]

[1]Harvard University,

[2]Massachusetts General Hospital,

[3]Franciscan Children's/ Boston Children's Hospital,

[4]Harvard T.H. Chan School of Public Health,

[5]Rutgers University,

[6]Wellesley College,

[7]MIT Media Lab

Abstract

Clinicians are tasked with predicting and preventing suicidal behavior among their patients; however, there is currently no method for accurately predicting whether a person will make a suicide attempt in the near future. We tested whether brief, smartphone-based surveys, combined with passively collected survey meta-data, could predict the occurrence of suicidal behavior over the next 7 days among those at elevated risk. Participants were 619 patients presenting to the hospital with suicidal thoughts/behavior. They were sent brief (20-item) smartphone-based surveys 6x/day for 3 months. Survey responses (N=79,448) and metadata (e.g., time since last survey submission) were used as predictors of next-week suicide attempt (SA) and suicide-related event (SRE; which also included hospitalization to prevent a SA) in a series of machine learning models. The most accurate prediction was achieved using bidirectional long short-term memory (BiLSTM) and simple lasso-penalized logistic regression models, with the best-performing model using BiLSTM to predict SRE, which with specificity at .90, had AUC=.94, sensitivity=.87, and positive predictive value=.30, and SAs with AUC=.90, sensitivity=.74, positive predictive value=.16. Prediction accuracy was higher than has been achieved in prior studies, and was strongest for models that predicted SREs (vs. SAs), included more sources of data, focused on adults (vs. adolescents), and when participants' own data were included in the model training process (vs. being held out). The strongest and most consistent predictors of next-week SA

Send correspondence to Matthew K. Nock, Harvard University, Cambridge, MA 02138; nock@wjh.harvard.edu.
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included within-study history of SREs (from adult lasso regression: OR=1.47) and self-reported agitation (OR=1.11), whereas odds of next-week SA were decreased for surveys submitted on weekends (OR=0.87) and in the context of feelings that one could resist suicidal urges (OR=0.88–0.96). Brief smartphone-based surveys can predict next-week SAs/SREs with a fairly high degree of accuracy. Future work is needed to further improve accuracy and test just-in-time interventions targeting high-risk periods.

General Scientific Summary

Suicide is a leading cause of death that has proven very difficult to predict, especially in the near-term (e.g., next week). We administered brief, smartphone-based surveys to a large sample of people experiencing suicidal thoughts and found that we could predict next-week suicide attempts with a fairly high degree of accuracy. The findings of this study may help clinicians identify which of their patients are at elevated risk of a suicide attempt in the next week so that additional preventive interventions can be provided to keep them safe.

Keywords

suicide attempt; suicidal; ecological momentary assessment; EMA; predict; smartphone

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Suicide is one of the leading causes of death and the primary form of mortality associated with mental disorders. Although significant advances have been made in the identification of risk factors for suicidal behavior (O'Connor & Nock, 2014; Moran et al., 2024; Nock & Wang, in press), a long-standing challenge to suicide prevention efforts is the inability to predict suicide attempts within time frames that can be acted upon by clinicians and family members, such as within the next few days or week (Glenn & Nock, 2014). Clinicians providing outpatient treatment, which most often involves weekly or monthly patient visits, currently have no method of determining whether a given patient will try to kill themselves between weekly visits.

Although there is significant debate about whether the use of smartphones is helpful or harmful to mental health (Haidt & Allen, 2020; Odgers, 2024), one positive feature is that they provide opportunities to measure important patient symptoms, such as suicidal thoughts and behaviors (STBs), in real-time with better temporal granularity than has previously been possible. Indeed, over the past 15 years, smartphones have been used to better understand the basic characteristics of suicidal thoughts including their frequency, duration, and severity (Nock et al., 2009; Coppersmith et al., 2023; Wang et al., 2024). However, to date, no studies have demonstrated an ability to accurately predict actual suicide attempts. Because suicide attempts are relatively rare in a given sample, predicting their occurrence requires relatively large samples of people at elevated risk followed for an extended period of time.

Moreover, most prior studies aimed at understanding and predicting STBs have measured hypothesized risk factors at only one or a small number of time points far in advance of these outcomes (Franklin et al., 2017). Rather than design this study as a test of one specific theory of STBs, we repeatedly assessed a range of key constructs hypothesized to predict the onset of STBs according to multiple leading theories of suicide (e.g., negative mood, hopelessness, burdensomeness, desire to escape) via smartphone surveys, providing an opportunity to test the extent to which such factors predict the near-term occurrence of STBs (Millner et al., 2020).

Here we report on the use of smartphone surveys that repeatedly assess a range of factors hypothesized to predict the short-term (next week) occurrence of suicide attempts. Overcoming limitations of prior studies, we used smartphone-based ecological momentary assessment (EMA) methods in a large sample (n>600) of people known to be at elevated risk of STBs (those with suicidal thoughts presenting for hospital-based psychiatric care) and followed them over an extended (3-month) time period.

Method

Sample

Participants (*N*=619) were recruited and enrolled from two Boston-area hospitals: 313 unique adults (18+ years) presenting to a psychiatric emergency service and 306 unique adolescents (12–19 years) presenting to a psychiatric inpatient unit. Inclusion criteria were: presenting with suicidal thoughts and/or recent suicidal behavior and owning an Android or iOS smartphone. Exclusion criteria were: any factor that impaired patients' ability to provide informed consent/assent, an inability to speak or write English fluently, presence of gross cognitive impairment due to florid psychosis, intellectual disability, dementia, acute intoxication, or the presence of extremely agitated or violent behavior. All participants provided written informed consent/assent for all study procedures. The study was approved by the institutional review boards of Harvard University (#IRB18–1749) and Mass General Brigham (#2019P001241).

Assessments

Participants provided informed consent, completed a baseline survey, and installed an app on their smartphone (LifeData) that delivered brief self-report surveys. Participants were compensated $10 for the baseline survey and $1 for each completed EMA survey. These data were then used to predict the occurrence of suicide attempts.

Surveys. Self-report smartphone surveys (20 items per survey) assessed participants' current/momentary experience of three aspects of suicidal thinking (urge, intent, and ability to resist suicidal urges) and 17 affective states (negative, hopeless, trapped, isolated, burdensome, angry, self-hate, agitated, worried, numb, fatigued, humiliated, desire to escape, desire to avoid, desire to approach, energetic, and positive). These items include key components of the leading theories of STBs (e.g., negative mood, hopelessness, budersomeness, isolation, trapped, desire to escape, etc.; Millner et al., 2020). For each item, the survey asked "Right now, how much do you feel…" followed by the item (accompanied by a brief definition of the term), which itself was followed by a 0–10 slider scale. See Supplemental Table S1 for the wording of each survey item. EMA surveys (assessing "right now") were sent to participants 6x/day for three months. Daily surveys (assessing "today") were delivered 1x/day for the next three months; however, given the current study focuses on EMA responses, we restrict our focus to data from the first three months of data collection.

Survey notifications began as soon as participants could access their phones, which typically was immediately for adults and after discharge from the inpatient units for adolescents. Notifications for the first and last surveys of the day were at fixed times set in collaboration with each participant at the time of enrollment (e.g., 9am, 9pm). Notifications for the middle four surveys were sent at randomized times, at least two hours apart, between the first and last surveys. Participants also could complete a user-initiated survey at any time to report the occurrence of a suicide attempt, non-suicidal self-injury, or another event that they thought would be important to inform us about. We used detailed risk monitoring procedures to respond in real-time to surveys indicating high suicidal intent (details available upon request).

Survey Metadata. We used several features from EMA survey metadata in the predictive models as such data can improve prediction accuracy for suicidal thoughts and behaviors (Zuromski et al., 2024). These included five characteristics of survey completion: (1) minutes lag from survey prompt to survey start, (2) minutes from survey start to survey submission, (3) survey completed but not submitted until next prompt, (4) days since last survey on file, and (5) number of extreme [0 or 10] responses. These also included six characteristics of survey start time: (1) daytime (i.e., survey started before sunset), (2) weekend, (3) holiday, (4) day before holiday, (5) minutes of daylight on that day, and (6) daylight curve (i.e., an inverted cosine curve with noon at the peak and earlier and later hours descending on either side).

Suicide Attempts and Suicide-Related Events. The primary outcome in this study was the occurrence of a suicide attempt (SA), which we defined as self-injurious behavior with some (non-zero) intention to die from that behavior. The secondary outcome was suicide-related events (SREs), a broader outcome that we defined as the presence of a suicide attempt OR suicidal behavior that stopped just short of a suicide attempt (i.e., aborted or interrupted suicide attempt), or presentation to a hospital with suicidal thoughts to prevent the occurrence of a suicide attempt. We included this broader outcome because these are also important clinical events whose prediction could enhance suicide prevention efforts. Suicide attempts and SREs were coded as present (0,1) if there was *either* (a) self-report of such an event in the EMA survey or (b) documentation of such an event in a clinical note in the electronic health record (consensus coded by two trained BA-level reviewers with supervision by a doctoral-level clinician with expertise in assessing/treating STB) or via a risk monitoring interaction. Given that the goal was longitudinal prediction of SAs/SREs, and that the occurrence of these events are among the strongest predictors of additional such events (Franklin et al., 2017), we also included three SRE features in the predictive models: (1) inpatient hospitalization during survey start date, (2) count of SREs since start of study period, and (3) days since last documented SRE.

Data Analysis

Time Scale of Prediction. The primary aim of this study was to use each participant's EMA data to prospectively predict whether they would have a SA/SRE within the next week (7 days) given responses to the 20-question EMA and metadata. We define "next week" as the window starting the date immediately after the survey in question, up to and including

6 days after the window begins. For example, if a survey is started on March 15th, the next week is defined as March 16th up to and including March 22nd.

Model Development and Validation. We developed two series of models, one predicting SAs and one predicting SREs. To avoid overfitting and to maximize the generalizability of results, we split our data into train (70%), validation (15%), and test (15%) sets. Training data were used to train the models, validation data were used to locate threshold values best approximating specificity=.90 for each feature subset, and test data were used to generate the estimated final performance of the model and threshold on each respective feature subset. Our model-selection strategy intentionally prioritized sensitivity over positive predictive value (PPV) since false positives are preferable to false negatives for suicide prevention.

We implemented two different approaches to splitting data into train, validation, and test sets: one in which *surveys* were randomly assigned to one of the three sets, and one in which *participants* were randomly assigned to one of the three sets. The former approximates situations in which an existing patient's data could be used to help train a model to predict their own behavior. The latter approximates situations in which a trained model could be used to predict SAs among new patients for whom no data had yet been collected. See Table S4 for participant and survey sample sizes in training, validation, and test sets for each model.

Model development started by examining a range of different prediction approaches, including lasso and ridge penalized logistic regression, K-nearest neighbors, XGBoost, random forest, and neural networks. These models were each instantiated with an exhaustive grid search, for which we attempted four data scalars: standard, robust, min-max, and none. Our logistic regressions also tried three class weight options: 0.9, 0.5, and none. Other applicable models saw variation in estimators and model-specific class weights. These combinations amount to 188 unique architectures comprised of 12 lassos, 12 ridges, 40 KNNs, 72 random forests, 4 XGBoosts, and 48 neural networks. Lasso with min-max scaling and 0.9 class weight for SA prediction and robust scaling and 0.9 class weight for SRE prediction consistently outperformed the other 186 options. Bidirectional Long Short-Term Memory (BiLSTM) was introduced later as a sequence-based (e.g., accounting for instances in which a sequence of surveys may generate a different prediction than singular survey-based risk levels), missing-enabled (e.g., considering missingness as a predictor) alternative to lasso regression. We then estimated model performance of our lassos and BiLSTMs among six primary feature sets using a Monte Carlo cross-validation (MC-CV) procedure. Model thresholds were selected in the validation phases to achieve a median 90% specificity.

We report the resulting sensitivity and positive predictive value (PPV) in the test sets. We repeated this process for 2 tasks: SA and SRE prediction, two split types: independent and random, two cohorts: adult and adolescent (based on recruitment site), and six different sets of predictive features: (1) EMA 17 affect items only, (2) EMA 3 suicide items only, (3) EMA all 20 items and (4–6) repeated sets 1–3 with the addition of metadata features.

See Supplemental Text for additional details about model development and validation procedures.

Results

Preliminary Analyses

Of the 619 unique participants consented, 502 (81.1%) provided data in at least one survey and those participants started a total of 79,448 surveys. Although participants started a large number of surveys, rolling median and mean survey initalization rates were <50% over the three-month study period and decreased over time (see Figure S1 in the Supplement). Over the 84 days (7 days * 12 weeks) of the study period, adult participants provided survey data for an average of 43 days (33.2 SD) and adolescents for 40 days (25.7 SD).

Depending on what questions and metadata are used, a minimum of 1,856 surveys and a maximum of 5,710 surveys were discontinued such that they became unusable for lasso (e.g., in models that used only the 3 SITB questions, only those 3 responses were required, whereas in models that used the affects, only those 17 responses were required). The final most inclusive sample included in the study analyses was 498 (246 adults, 252 adolescents) unique participants (80.5% of those recruited) who completed the affect questions on at least one survey. These participants most inclusively answered required questions on 77,592 surveys. Sample characteristics of those included in study analyses are reported in Table 1.

Over the course of the study, we observed 459 SAs/SREs among 193 unique participants. Restricting these to only participants who provided any survey data reduced these counts to 372 events among 162 participants. A match is only possible in instances for which we had any survey from the participant in the prior 7-day window, which generated 170 event matches – 85 SAs and 85 (non-SA) SREs – across 92 unique participants. SREs (inclusive of SAs) occurred in the prediction window of 766 out of 37,998 adolescent surveys (2.02%) and 2,252 out of 41,450 adult surveys (5.43%). SAs occurred in the prediction window of 516 adolescent surveys (1.36%) and 974 adult surveys (2.35%). These prevalence rates are based on the number of started EMAs, which do not always allocate to model training.

Model Performance

Prediction of Suicide Attempts. Model performance (AUC) was consistently stronger when using BiLSTM and lasso regression than all other modeling approaches examined, so we focused on those approaches in all subsequent analyses. For both approaches, and among both adults and adolescents, performance was consistently stronger when surveys were randomly split across train, validation, and test (“random;” test AUCs= .67–.90) than when participants were held out for independent validation (“independent;” test AUCs= .36–.77), as revealed both by the lower point estimates and greater error in the latter; Figure 1). Performance also generally improved when more data were added to the model. These patterns also were observed when examining other model metrics such as sensitivity and PPV (Figure 2).

The best predicting model using a random data split was a BiLSTM that used all survey items and metadata, which for adults achieved AUC=.90, sensitivity=.74, PPV=.16 and for adolescents: AUC=.87, sensitivity=.65, PPV=.06 (see Figure 2). The best predicting model using an independent data split was a lasso-penalized logistic regression (with a 0.9 minority class weight) that used all survey items and metadata for adults: AUC=.75, sensitivity=.44, PPV=.08 and all survey items (but no metadata) for adolescents: AUC=.67, sensitivity=.39, PPV=.03. Both models for the SA prediction task use min-max scaling.

Prediction of Suicide-Related Events. Prediction accuracy was even stronger when predicting SREs. Performance again was consistently stronger when surveys were randomly split across train, validation, and test (AUCs= .70–.94) than when participants were held out for independent validation (AUCs= .32–.80; see Figure S2). The best predicting model using a random data split was BiLSTM that used all survey items and metadata, which for adults achieved: AUC=.94, sensitivity=.87, PPV=.30 and for adolescents: AUC=.94, sensitivity=.86, PPV=.11 (see Figure S3 for all model metrics). The best predicting model using an independent data split was a lasso-penalized logistic regression that used only the 3 SITB items and metadata for both adults: AUC=.80, sensitivity=.53, PPV=.21 and adolescents: AUC=.72, sensitivity=.29, PPV=.03.

Analysis of False Positives. Although the rate of true positives increased when expanding the focus from SAs to SREs (e.g., adult lasso PPV increased from .08 to .19), the rate of false positives remained quite high. As such, we examined the characteristics of our false positive cases to inform both the clinical utility of our current models and future prediction studies. Most false positives (50%) were characterized by participants' self-reporting elevated scores on affective states and/or suicide-related items. For instance, for 50% of SA false positives, participants endorsed at least a 7/10 on 11/17 affect items (see Figure S5) and/or 1 suicide-related items (Figure S6) – indicating a level of distress that may benefit from clinical intervention. If these instances of elevated distress are considered part of the outcome definition (i.e., outcome = SA + SRE + significant distress), our PPV increases to .77 for adults and .68 for adolescents on average. This means that when the model indicates a person is at elevated risk, 68–77% of the time there is a clinically concerning event in the next 7 days.

Feature Importance

We examined the associations between each of the features included in the models predicting next-week SAs and SREs. Unscaled coefficients from MC-CV runs for each independent lasso model are presented for SAs in Figure 3 and for SREs in Figure S4. Specific corresponding odds ratios are presented in Table 2 for both adults and adolescents. There is general consistency in results for feature importance across modeling approaches and adult/adolescent samples. For instance, focusing on the prediction of SAs and regardless of which approach was used, the largest increase in odds of a SA in the next week is associated with more SREs during the study period, higher self-reported agitation, the presence of extreme scores (i.e., number of negative items endorsed as "10/10" or positive items endorsed as "0/10"), and increased self-reported negative affect, numbness, suicidal urge, desire to avoid, and anger (Table 2). The largest and most consistent decreases in odds are associated with surveys started on a weekend, elevated ability to resist suicidal urges, more days since last known SRE, and feeling worried. Results for SREs are presented in Table S5.

Most features showed consistency across all models for SA (Figure 3) and SRE (Figure S4), with a few notable differences. The largest observed differences were for self-reported agitation, extreme scores, and desire to avoid, which were all among the strongest predictors of SA but the weakest predictors of SREs. In contrast, features that were among the strongest predictors of SRE but not SA included surveys started on a holiday, the day before a holiday, and surveys with elevated suicidal intent.

Discussion

There are four key findings from this study. First, data from smartphone-based surveys can predict next-week SAs/SREs risk with a high degree of accuracy. Second, accuracy was strongest for models predicting SREs (vs. SAs), including more sources of data, for adults (vs. adolescents), and when participants' own data were included in (vs. excluded from) the model training process. Third, although all models had a high false positive rate, approximately half (50%) of our false positive cases included self-reported high negative affect and/or suicidal thoughts. Fourth, results revealed several features that consistently predicted SAs/SREs, including extreme scores and high negative affect (especially agitation). Each of these findings warrants additional comment.

The most important finding from this study is that we were able to use brief surveys and associated data from people's smartphones to predict the occurrence of a next-week SA or SRE with a fairly high degree of accuracy. The accuracy of our best prediction model (AUC=.94) compared favorably with results reported in studies using much wider prediction windows (e.g., months and years; Nock et al., 2022; Bayramli et al., 2022; Barak-Cohen et al., 2020) and our PPVs (SA=16%, SRE=30%) were higher than what has been found in prior studies with these larger time windows (2–4% for SAs; Belsher et al., 2019; Kessler, 2019; Simon et al., 2019). The sensitivity of our models (up to 87%) indicated that our prediction models were capturing a large percentage of SAs and SREs up to 1 week before they occurred. These results, which were cross-validated and robust across modeling approaches and samples, demonstrate that we can reliably predict SAs/SREs in the days before they occur, supporting the potential development and deployment of just-in-time interventions (Coppersmith et al., 2022). Future efforts will be aimed at further increasing predictive accuracy to capture more instances of SA (sensitivity) and maximize the rate of true positives (PPV). In the meantime, efforts should be undertaken to study the implementation of this approach in clinical settings to test its ability to prevent SAs among those at risk for such outcomes.

Notably, the accuracy of our predictions varied in several important ways that can inform future scientific and clinical efforts. First, model accuracy generally improved when metadata from the surveys were included. This result highlights the value of collecting and incorporating data beyond self-report in future prediction efforts. Second, model accuracy was consistently much stronger when participants' own data were allowed to be included in model training than when data were trained using data from some participants and applied to a group of independent participants. This is not surprising, and an expected outcome of these different approaches to modeling the data. Nevertheless, the clear difference in results for predicting SAs and SREs raises an important question for researchers and clinicians to consider: in clinical settings, when seeing a new patient, should we collect data from them and train models with their data or should we apply existing models to new patients as soon as they enter the clinic? The former approach will lead to more accurate predictions, but only after some period of new data collection, whereas the latter approach will allow us to make more immediate predictions about patient risk of SA/SRE, but with less reliability and detection capability. Recent advances in this area have highlighted that it is possible to build unique models for each patient (Wang et al., 2024), but that even greater accuracy can be achieved, and more quickly, by leveraging data from similar patients (Hang et al., 2025). Determining which approach achieves the greatest clinical utility in real-world settings is an important issue for clinical implementation efforts. Third, although we achieved fairly strong prediction for both adult and adolescent models, results generally were stronger among adults. Prediction was especially weak in the case of independent models for adolescents. This finding highlights the importance of considering adolescent-specific models and suggests that it may be especially challenging to predict SAs/SREs without incorporating data from the adolescent whose behavior we are trying to predict. Fourth, our prediction metrics were consistently stronger for models predicting SREs than SAs. This is likely due, at least in part, to the higher prevalence of SREs. Our ability to predict actual SAs and not just the occurrence or severity of suicidal thinking as has been done in prior studies (Wang et al., 2024; Oquendo et al., 2024; Choo et al., 2024) is a strength of this study. But the prediction of SREs is important in itself as it would be helpful for clinicians and parents to know in advance when higher levels of care are needed to prevent the suicidal crisis.

One of the biggest critiques of prediction models for SAs is their high false positive rate (low PPV). Studies focused on predicting the 12-month occurrence of SAs among patients typically achieve PPVs of 1–4% (Belsher et al., 2019). Some argue these rates are too low to be clinically useful (Belsher et al., 2019) whereas others note that these rates are aligned with those from other areas of medicine that successfully guide intervention efforts (e.g., heart disease, cancer; Kessler, 2019; Simon et al., 2019) and that acting on models with this level of accuracy is cost-effective (Ross et al., 2021). Two findings from the current study are especially relevant in this regard. First, as noted above the PPVs achieved in this study are higher than those achieved in prior studies (up to 30% in the current study), and were achieved with much shorter time windows, facilitating the deployment of intervention efforts aimed at preventing SAs/SREs. Second, we found that half (~50%) of our false positive cases were instances in which, although there was not a SA/SRE, there was an important increase in negative affect and/or suicidal thoughts that might benefit from intervention. This further supports the use of this smartphone monitoring approach for clinical intervention.

The prediction performance we observed was driven by several key features that were associated with significant increases and decreases in risk. First, results revealed that each new occurrence of a SRE during the post-hospital monitoring period of adults was associated with a 47% increase in the odds of a SA, and 65% increase in the odds of another SRE. It is well-established that SAs/ SREs are associated with increased risk of additional such behaviors (Franklin et al., 2017; Ribeiro et al., 2016). The current results indicate that this finding is true in the short-term as well, which is highly relevant to the clinical prediction of suicidal behaviors. Second, self-reported negative affect, and especially the feeling of agitation, significantly predicted SA/SREs. Each additional point endorsed for “agitation” on a 0–10 scale was associated with an 11% increase in the odds of making a SA in the next week among adults and 6% increase among adolescents. Agitation is an important, but often underappreciated, correlate of suicidal behavior (Busch et al., 2003) and the current results document its importance in the short-term prediction of SAs. Third, each of the three suicide-related items showed good predictive validity. These findings support the use of these items in the clinical assessment of short-term risk of suicidal behavior. Although assessment of suicidal urge and intent are common in suicide risk assessment (Ward-Ciesielski & Wilks, 2020), ability to resist suicidal urges is not and our results suggest that it should be included in such assessments, especially given converging findings suggesting that this construct is predictive of future SA (Rogers et al., 2024). Future work should test the targeting of these and other features in preventive interventions. Fourth, results also revealed that inpatient hospitalization was associated with decreased odds of next-week SAs/SREs. This finding is important given ongoing debate about the therapeutic value, or potential iatrogenic effects, of hospitalization for suicide risk (Ross et al., 2024; Large & Kapur, 2018). Several other findings emerged that should be considered exploratory and followed up in future studies. The day before a holiday was associated with decreased risk of SA within the next week, contributing to prior evidence of a small decrease in risk on many holidays (Lee et al., 2024). Fifth, the same feature set that best-predicted SAs also best-predicted SREs; thus, the same model could be used clinically to predict both outcomes. Ideally, future work will provide the field with an ability to predict increasingly fine-grained outcomes, such as informing clinicians when a given patient will experience suicidal thinking, severe suicidal thinking, and transition from suicidal thinking to a suicide attempt. However, we have not yet achieved that level of precision. Sixth and finally, the fact that a number of items were not significantly associated with SA/SRE also is helpful for future efforts, as translating these results to clinical use will likely require using a smaller item set. The use of briefer surveys also may translate into higher levels of engagement (Smyth et al., 2021; Eisele et al., 2022), which could improve model reliability and detection and prevention capabilities.

Limitations

These results should be interpreted in the context of four key limitations. First, our sample included only those patients with smartphones and willing to complete smartphone surveys following their presentation to a hospital. Other aspects of our recruitment also likely limited the generality of our obtained results (e.g., all data were collected in the Northeast US, there were differences in how adults and adolescents were recruited – and overlap in age between the two samples).

Second, there was a significant amount of missing data, even in the context of our paying participants for completing surveys, limiting the ecological validity of our findings and highlighting that the intensive longitudinal monitoring methods used will not be equally engaging to all participants. On balance, despite this level of missingness, we still had nearly 80,000 EMA datapoints. With those data, we were able to sufficiently engage most of this high-risk sample and to sustain their engagement long enough to build prediction models for this high-risk post-hospital period, supporting the feasibility of this approach. Notably, these rates of participation were obtained using our 20-item protocol, and future efforts will be helped by the fact that much of our predictive power came from a smaller set of features. Our predictive models dealt differently with missingness, which may help to explain why BiLSTM generally outperformed lasso models (i.e., BiLSTM incorporates information about missingness, whereas lasso does not). Prior studies suggest that the prediction of suicidal behavior can be improved when missingness is included in prediction models (e.g., Wang et al., 2021) – and we plan to explore this topic in greater depth in or future work.

Third, our sensitivity and PPV were relatively modest, and we failed to predict many of the SAs/SREs that occurred during the study period, and most of our predictions were false positives. Given our response variable depends on accurate reporting of SAs and SREs by the participating hospital network or the participant themself, we cannot know if any of our denoted false positives are actually true positives. Moreover, our PPVs increase to 68–77% if we include in our outcome definition episodes of extremely high negative affect and suicidal thinking – suggesting that much of the time our “false positives” are still capturing instances in which it would be worth triggering a low-cost intervention aimed out reaching out and helping the participant/patient. Further improving the reliability and detection capabilities of our predictive models is a primary focus for future work in this area.

Fourth and related, our models included a relatively limited set of predictors. Future research must examine a broader array of potential risk factors including passively observed physiological and behavioral (e.g., heart rate, accelerometry), social/interpersonal (e.g., call and text logs), and environmental (e.g., stressful events) factors to facilitate the improved prediction and prevention of suicidal behavior.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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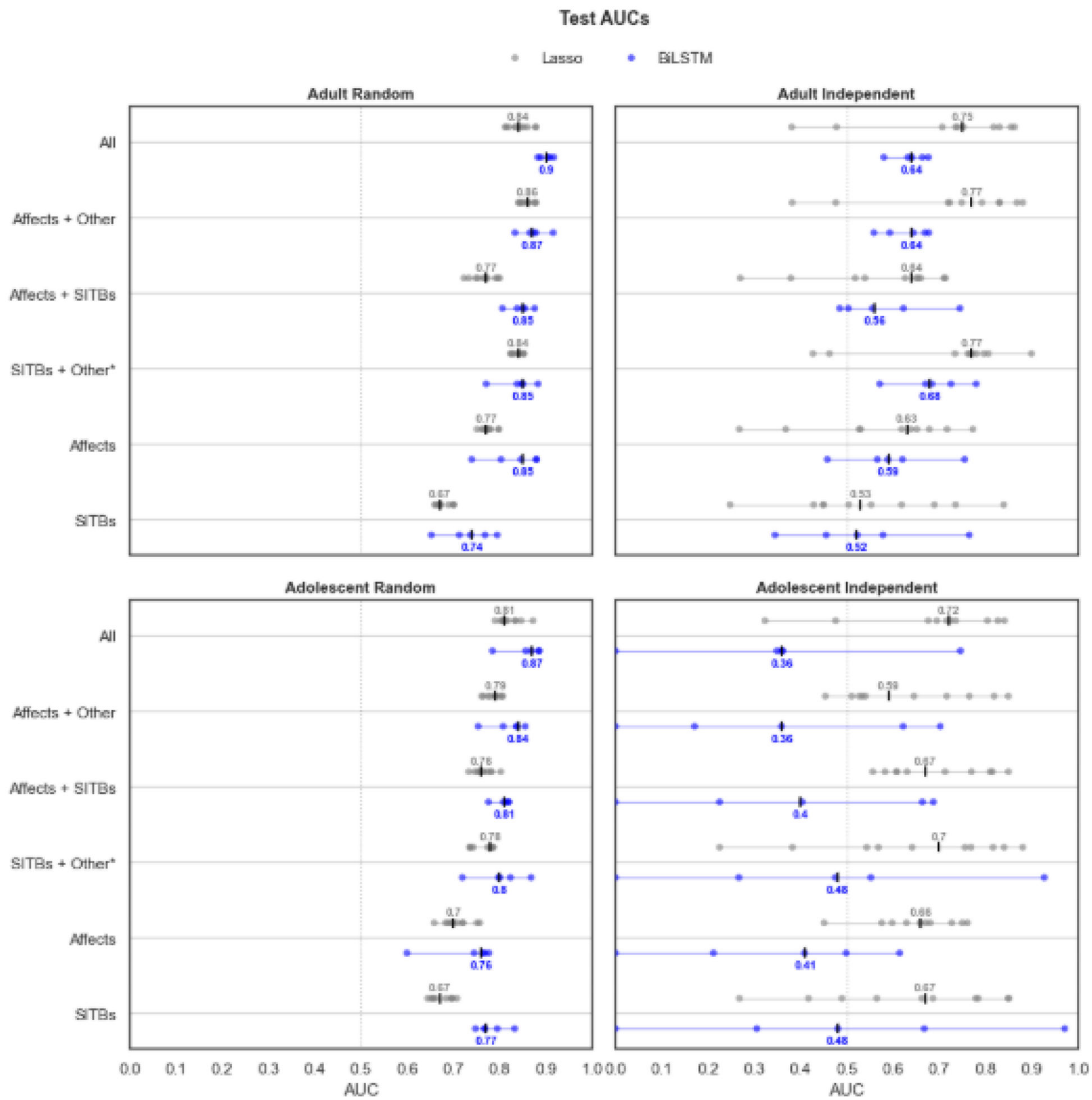

Figure 1. AUCs for Predicting Suicide Attempts

Note: AUC = Area under the curve; BiLSTM – Bidirectional Long Short-Term Memory.

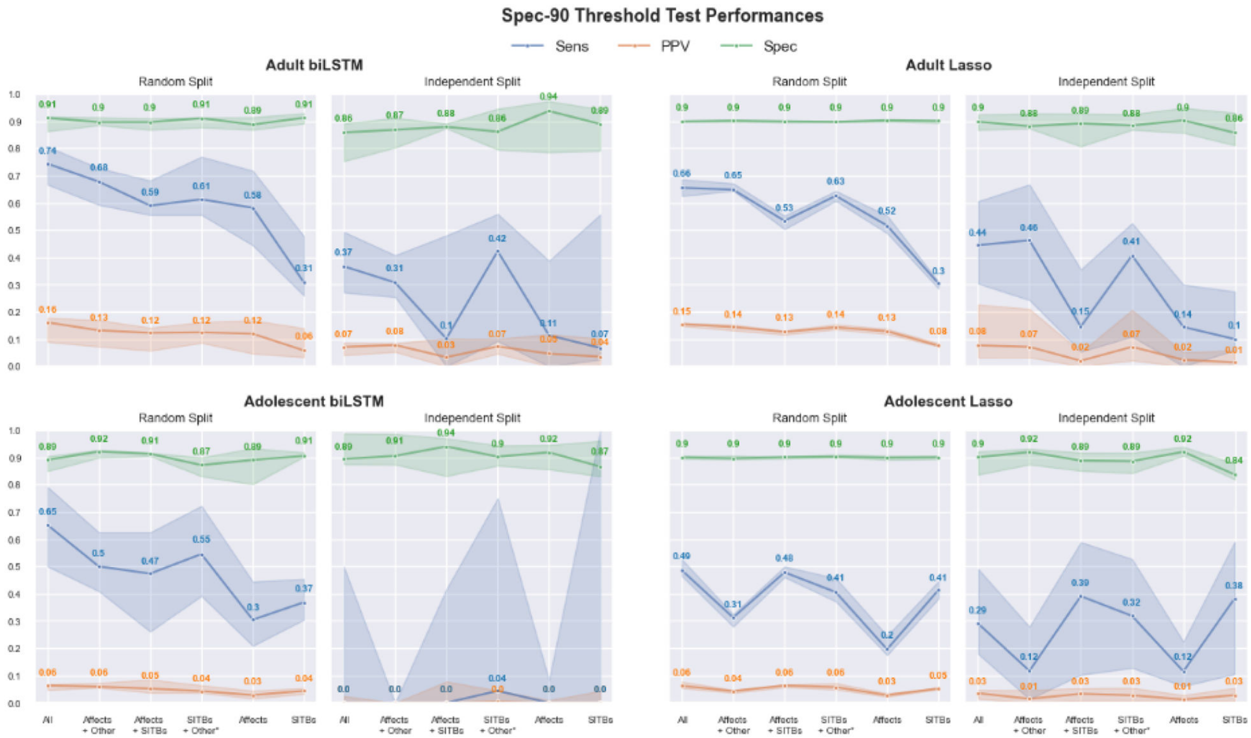

Figure 2. Accuracy Metrics for Models Predicting Suicide Attempts

Note: Details of the process for generating these values can be found in the supplemental text under the Modeling section. CI = Confidence interval; Val = Validation; SITBs = Self-injurious thoughts and behaviors; PPV = Positive predictive value; Sens = Sensitivity; Spec = Specificity.

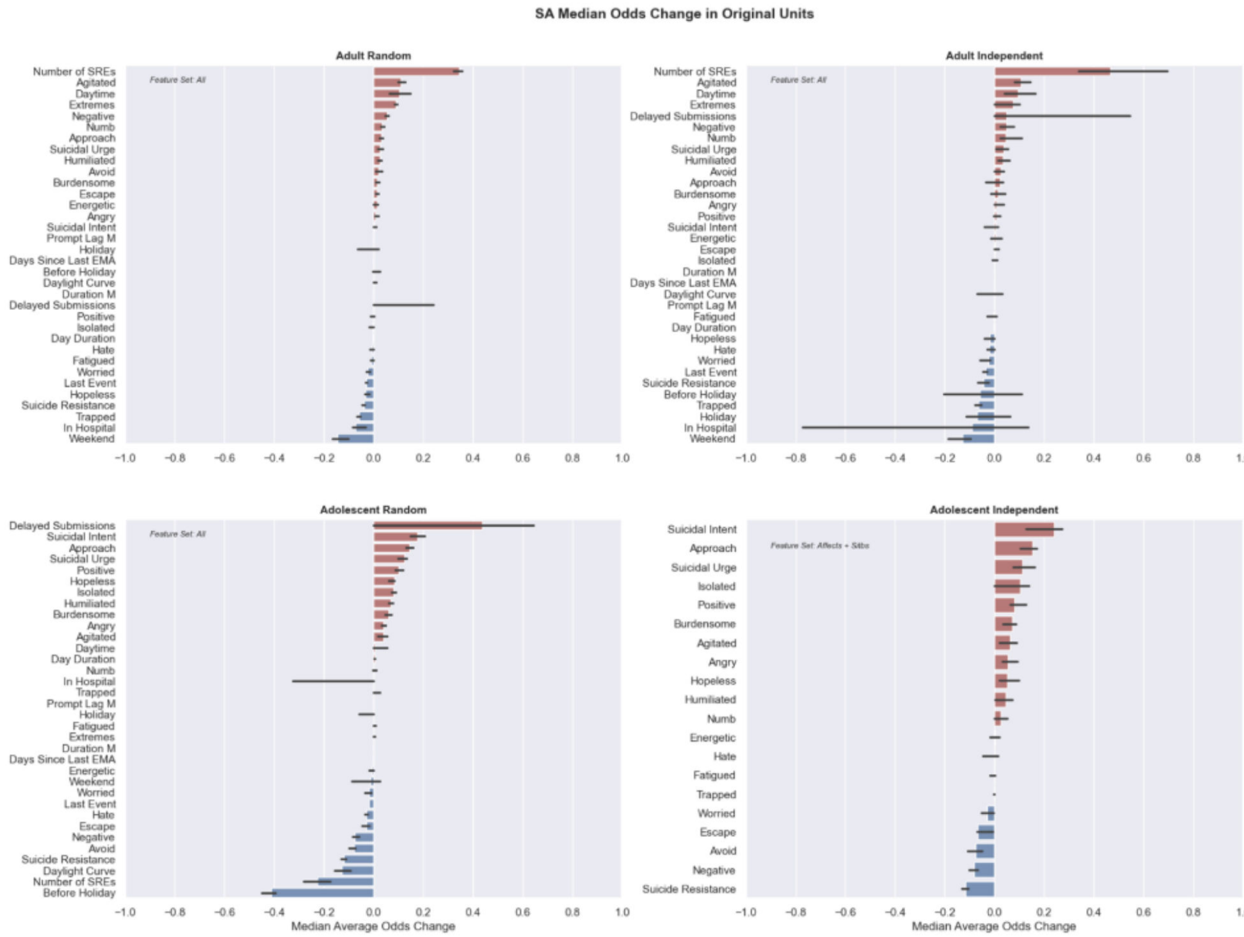

Figure 3. Features Predicting Next-Week Suicide Attempt

Note: Chosen feature set odds ratios from lasso results. The visualized feature importance depends on the chosen feature set and is learned based on the data in the training set. Odd ratios are shown in their original units, rather than their scaled ones. Error bars report the median at a 95% confidence interval. SA = Suicide attempt; SRE = Suicide-related event; M = Minute; EMA = Ecological momentary assessment.

Table 1.

Sample Characteristics (*n*=498)

Variable	%	N	M	SD
Age (12–69 years)			23.1	18.0
Race				
White	74.5	371		
Multiracial	3.4	17		
Black	7.6	38		
Asian	5.6	28		
Missing	8.8	44		
Ethnicity				
Non-Hispanic	82.3	410		
Hispanic	14.7	73		
Missing	3.0	15		
Sex At Birth				
Female	72.1	359		
Male	27.7	138		
Missing	0.2	1		
Income				
$0 – 20,999	24.1	120		
$21,000 – $40,999	9.6	48		
$41,000 – $60,999	4.4	22		
$61,000 – $80,999	2.6	13		
$81,000 – $100,000	1.8	9		
>$100,000	2.4	12		
Missing	4.4	22		
Not Applicable (Adolescents)	50.6	252		
Lifetime suicidal ideation *	89.4	445		
Among those with lifetime ideation				
Past month suicidal ideation	92.8	413		
Lifetime suicide plan	24.0	107		
Past month suicide plan	67.4	300		
Lifetime suicide attempt	32.1	160		
Past month suicide attempt	29.1	145		

* **Note:** All participants had suicidal thoughts as part of their clinical presentation, this variable and section refer to endorsement of suicidal thoughts/plan/attempt on our baseline self-report assessment.

Table 2.

Features Predicting Suicide Attempts (Independent Lasso)

Category	Feature	OR Interpretation	OR (95-CI) Adult (All)	OR (95-CI) Adolescent (Affects + SITBs)
Other	Number of SREs	For every additional event experienced since study enrollment	1.47 (1.37 – 1.63) **	-
Affect	Agitated	For every additional point on a 0 – 10 scale	1.11 (1.08 – 1.15) **	1.06 (1.01 – 1.09) *
Other	Daytime	If the survey was submitted during the daytime	1.10 (1.05 – 1.16) **	-
Other	Extremes	For every additional negative survey question marked as a 10 or positive survey question marked as a 0	1.08 (1.02 – 1.11) **	-
Other	Delayed submissions	If the survey was submitted at the next survey prompt (forgotten)	1.05 (0.99 – 1.52)	-
Affect	Negative	For every additional point on a 0 – 10 scale	1.05 (1.03 – 1.07) **	0.92 (0.9 – 0.93) **
Affect	Numb	For every additional point on a 0 – 10 scale	1.05 (1.03 – 1.09) **	1.03 (1.01 – 1.05) *
SITB-SI	Suicidal urge	For every additional point on a 0 – 10 scale	1.04 (1.0 – 1.06) *	1.11 (1.08 – 1.16) **
Affect	Humiliated	For every additional point on a 0 – 10 scale	1.04 (1.02 – 1.05) **	1.05 (1.02 – 1.07)
Affect	Desire to avoid	For every additional point on a 0 – 10 scale	1.03 (1.01 – 1.04) **	0.92 (0.9 – 0.95) **
Affect	Desire to approach	For every additional point on a 0 – 10 scale	1.02 (0.97 – 1.06)	1.15 (1.11 – 1.17) **
Affect	Burdensome	For every additional point on a 0 – 10 scale	1.01 (0.99 – 1.04)	1.07 (1.04 – 1.09) **
Affect	Angry	For every additional point on a 0 – 10 scale	1.01 (1.0 – 1.03) *	1.06 (1.02 – 1.1) **
Affect	Positive	For every additional point on a 0 – 10 scale	1.01 (0.98 – 1.03)	1.08 (1.06 – 1.12) **
SITB-SI	Suicidal intent	For every additional point on a 0 – 10 scale	1.01 (0.97 – 1.02)	1.24 (1.13 – 1.28) **
Affect	Energetic	For every additional point on a 0 – 10 scale	1.00 (0.99 – 1.03)	1.00 (0.98 – 1.03)
Affect	Desire to escape	For every additional point on a 0 – 10 scale	1.00 (1.0 – 1.02)	0.93 (0.93 – 0.98) **
Other	Days since last EMA	For every additional day since the last EMA submission	1.00 (1.0 – 1.0)	-
Other	Duration (minutes)	For every additional minute the survey takes	1.00 (1.0 – 1.0)	-
Other	Prompt lag (minutes)	For every additional minute it takes to open the survey from prompt notification	1.00 (1.0 – 1.0)	-
Other	Daylight curve	As survey start approaches peak environmental alertness (noon)	1.00 (0.94 – 1.04)	-
Affect	Isolated	For every additional point on a 0 – 10 scale	1.00 (0.98 – 1.02)	1.10 (1.03 – 1.13) **
Affect	Fatigue	For every additional point on a 0 – 10 scale	1.00 (0.98 – 1.01)	1.00 (0.97 – 1.02)
Other	Day duration	Minutes of daylight (Sunset – Sunrise)	1.00 (1.0 – 1.0) **	-
Affect	Hopeless	For every additional point on a 0 – 10 scale	0.98 (0.96 – 0.99) *	1.05 (1.03 – 1.09) **
Affect	Self hate	For every additional point on a 0 – 10 scale	0.98 (0.97 – 1.0) *	1.00 (0.96 – 1.02)
Affect	Worried	For every additional point on a 0 – 10 scale	0.98 (0.95 – 0.98) **	0.97 (0.95 – 0.99) *
Other	Days since last event	For every additional day gone without an SRE or SA	0.97 (0.96 – 0.97) **	-
SITB-SI	Suicide resistance	For every additional point on a 0 – 10 scale	0.96 (0.94 – 0.97) **	0.88 (0.87 – 0.9) **
Other	Before holiday	If the survey was done the day before a holiday	0.94 (0.83 – 1.09)	-
Affect	Trapped	For every additional point on a 0 – 10 scale	0.94 (0.92 – 0.95) **	1.00 (1.0 – 1.0)

Category	Feature	OR Interpretation	OR (95-CI) Adult (All)	OR (95-CI) Adolescent (Affects + SITBs)
Other	Holiday	If the survey was done on a holiday	0.93 (0.88 – 1.04)	-
Other	In hospital	If the survey was done during a known hospitalization	0.91 (0.41 – 1.06)	-
Other	Weekend	If the survey was submitted on a Saturday or Sunday	0.87 (0.81 – 0.92) **	-

Note: OR results from two-sided 1 sample t-test with population mean set to 1 (odds ratio = no change) from scipy.

** indicates $p < 0.01$ and

* indicates $p < 0.05$. OR = Odds Ratio. 95-CI = 95% Confidence Interval. Point Estimate reported is the median.